

Attrition Prediction & Strategy

Reducing IBM staff turnover via targeted high-risk employee

[GitHub Link](#)

| Executive Summary

The Challenge

IBM's current attrition rate stands at 16.12%, creating a critical need for intervention.

Primary Drivers

Overtime, Salary, and Job Level are the primary factors influencing employee turnover.

Strategic Goal

Reduce the overall attrition rate to below 10% through data-driven retention.

The Result

Successfully reduced projected attrition to 7.6%.

Methodology



01. Define

Problem Statement, establish project background, business objectives.



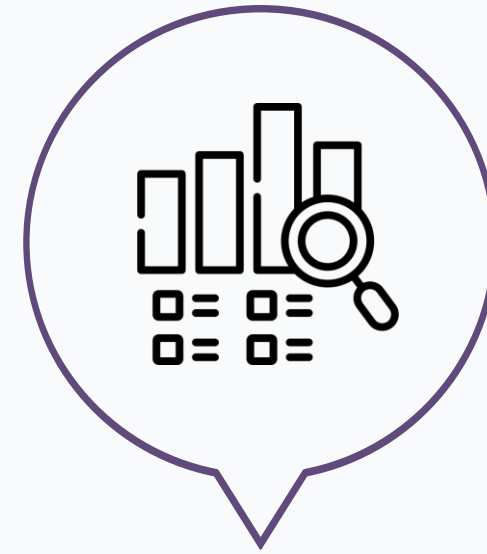
02. Prepare

Data Cleaning, process raw data using Python to handle missing values and noise.



03. Explore

Data Analysis, perform exploratory analysis to gather initial patterns and information.



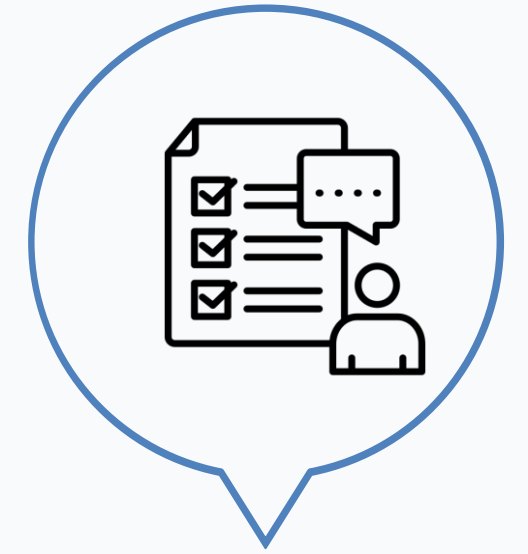
04. Analyze

Apply logistic regression to predict attrition and perform impact analysis.



05. Visualize

Build Power BI dashboards for key findings.

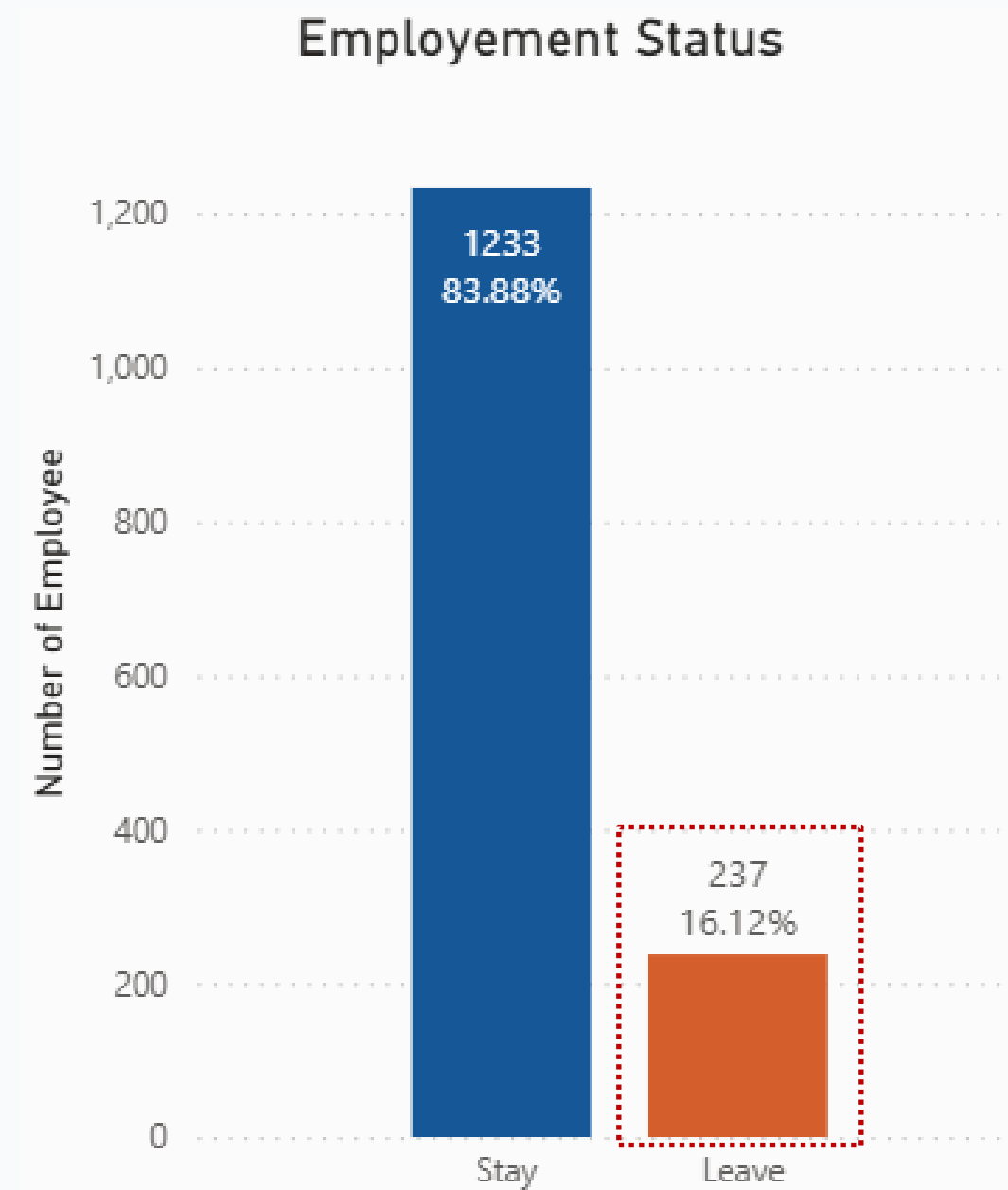


06. Recommend

Translate analysis into business recommendations.

Defining Business Problem

| Project Background

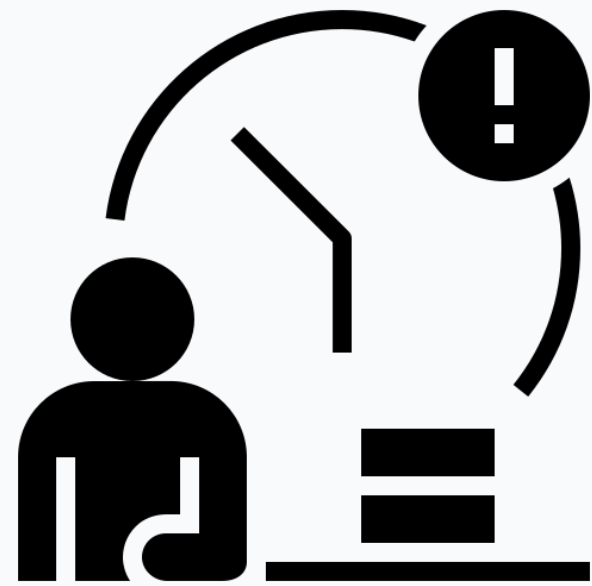


With **237 out of 1,470** employees leaving, IBM is experiencing a high attrition rate of **16.12%**.

This project objective is to identify key turnover drivers and implement strategies to reduce this rate.

| 3 Key Primary Factors

Overtime, Salary, and Job Level has most impact on. Attrition.



53.59% employee who left the company worked overtime



68.78% employee who left the company have salary less than \$5,000



82.28% employee who left the company are Juniors

| Project Goal



Our project goal is to reduce IBM's attrition rate to **below 10%** by implementing targeted retention strategies for **high-risk employees** identified by frequent **overtime, low salaries (under \$5,000), and junior job levels.**



| Data Cleaning



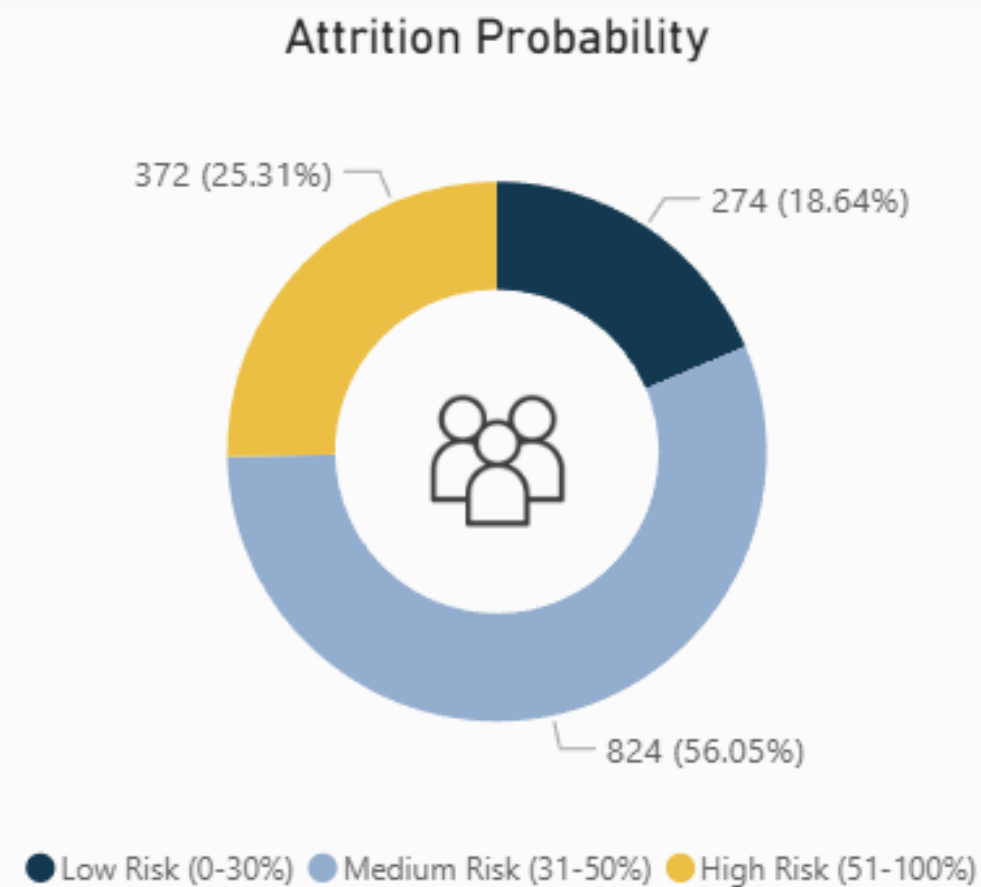
For a detailed walkthrough of the data cleaning process, please refer to the complete Python codebase in the project repository: [Link](#)



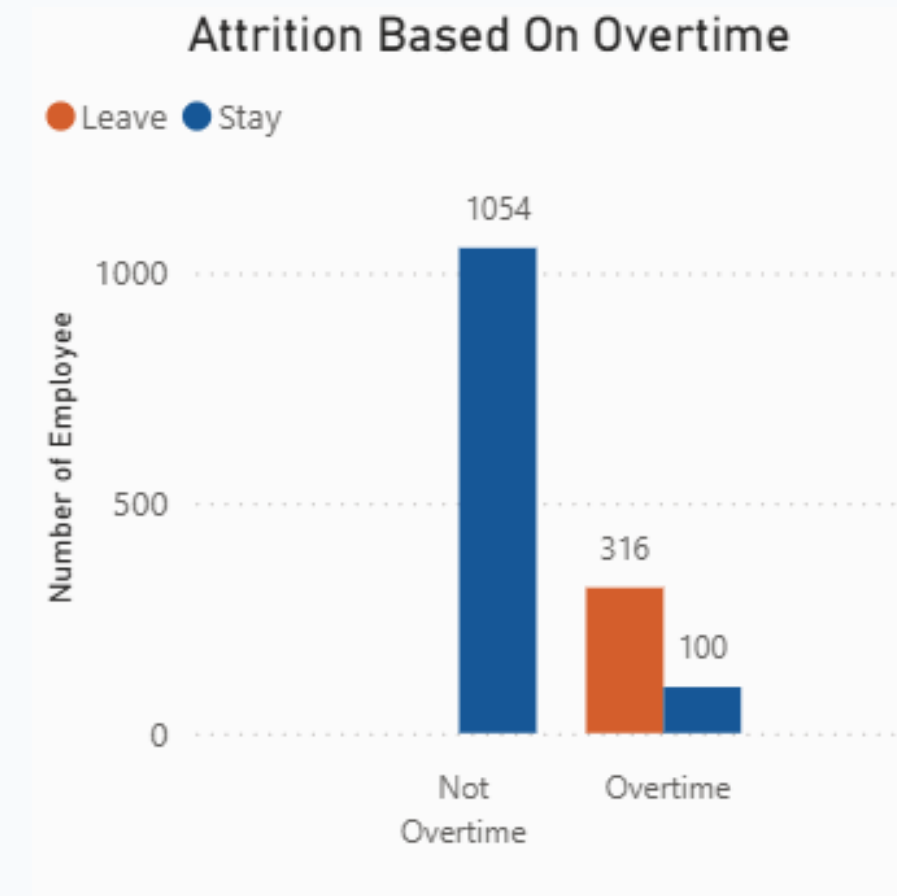
Data Analysis

The prediction model is based on Logistic Regression.

Total employee: 1470



- With **44%** of our workforce in the high-to-mid risk zones, this distribution shows where we must prioritize immediate retention resources.



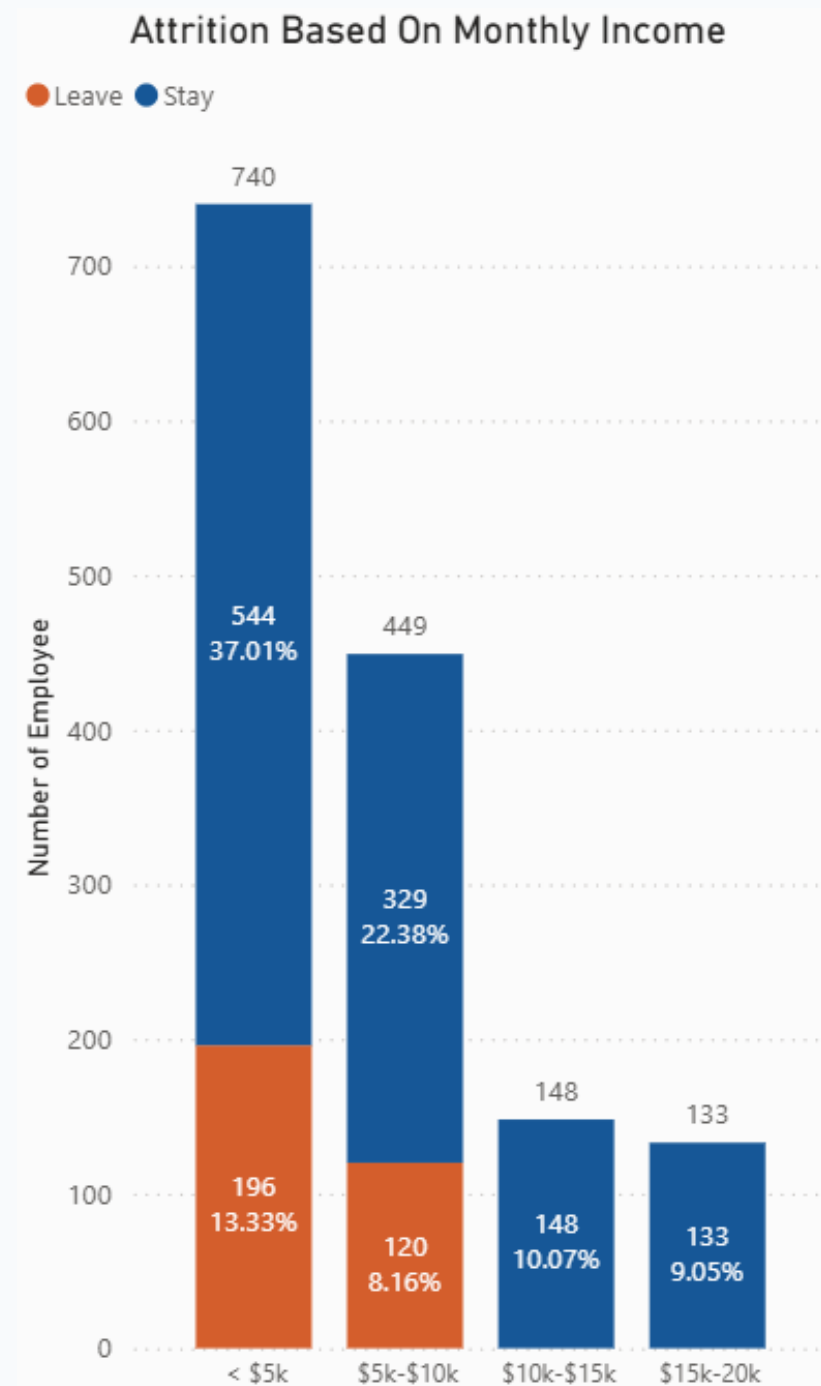
- Analysis reveals a critical correlation between overtime and attrition.
- 100% of employees predicted to leave in the model fall within the overtime group.

- ✓ Overtime tends to lead burnout and lead employee attrition.
- ✓ We can restrict overtime policy for the new strategy.

Data Analysis

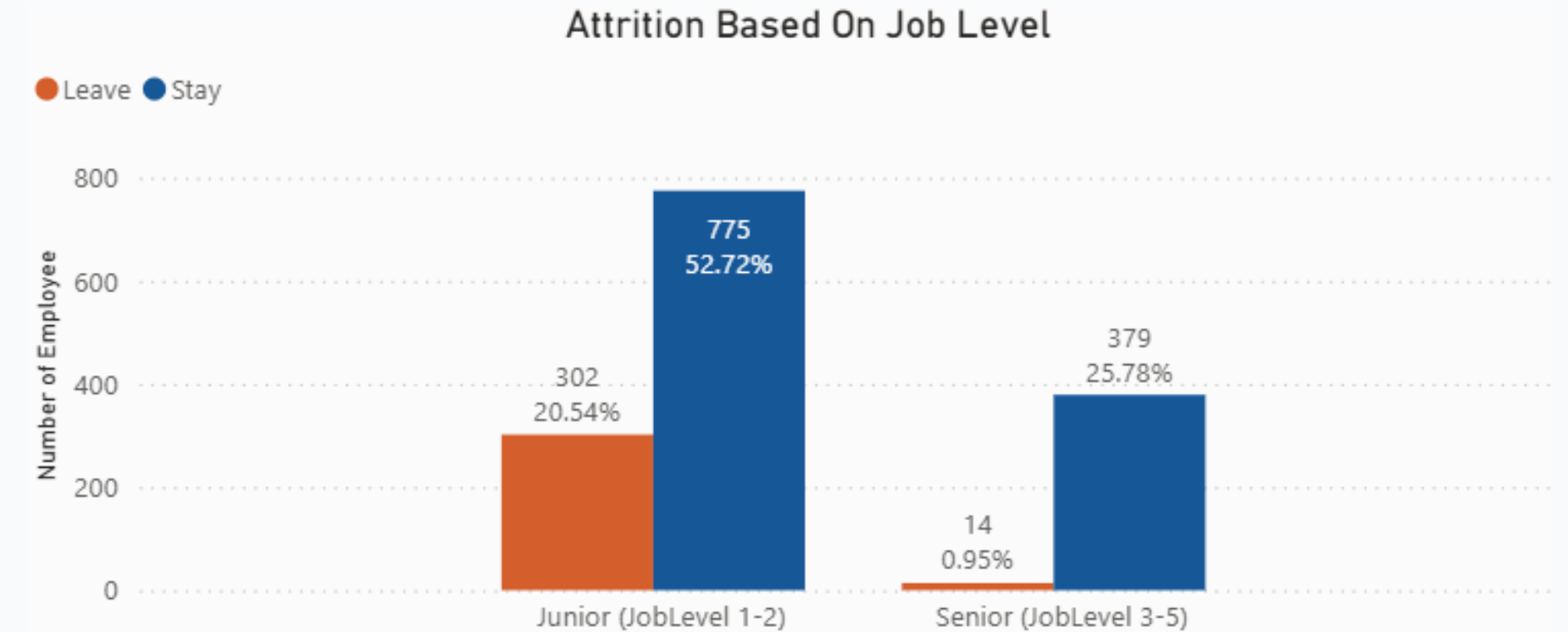
The prediction model is based on Logistic Regression.

Total employee: 1470



- Analysis indicates that attrition is concentrated among lower-income brackets, as 316 employees predicted to leave earn less than \$10,000 per month.
- Out of 316, 196 employees predicted to leave earn less than \$5,000 per month.

✓ Suggesting a need for performance-based compensation adjustments.



- The predictive model identifies that junior-level staff are at a significantly higher risk of attrition.
- 302 junior workers predicted to leave compared to only 14 senior-level employees, highlighting the need for targeted promotion and retention strategies for early-career staff.

Strategy

Strategy

Targeting high risk attrition employees (Overtime, low salary, and junior level)

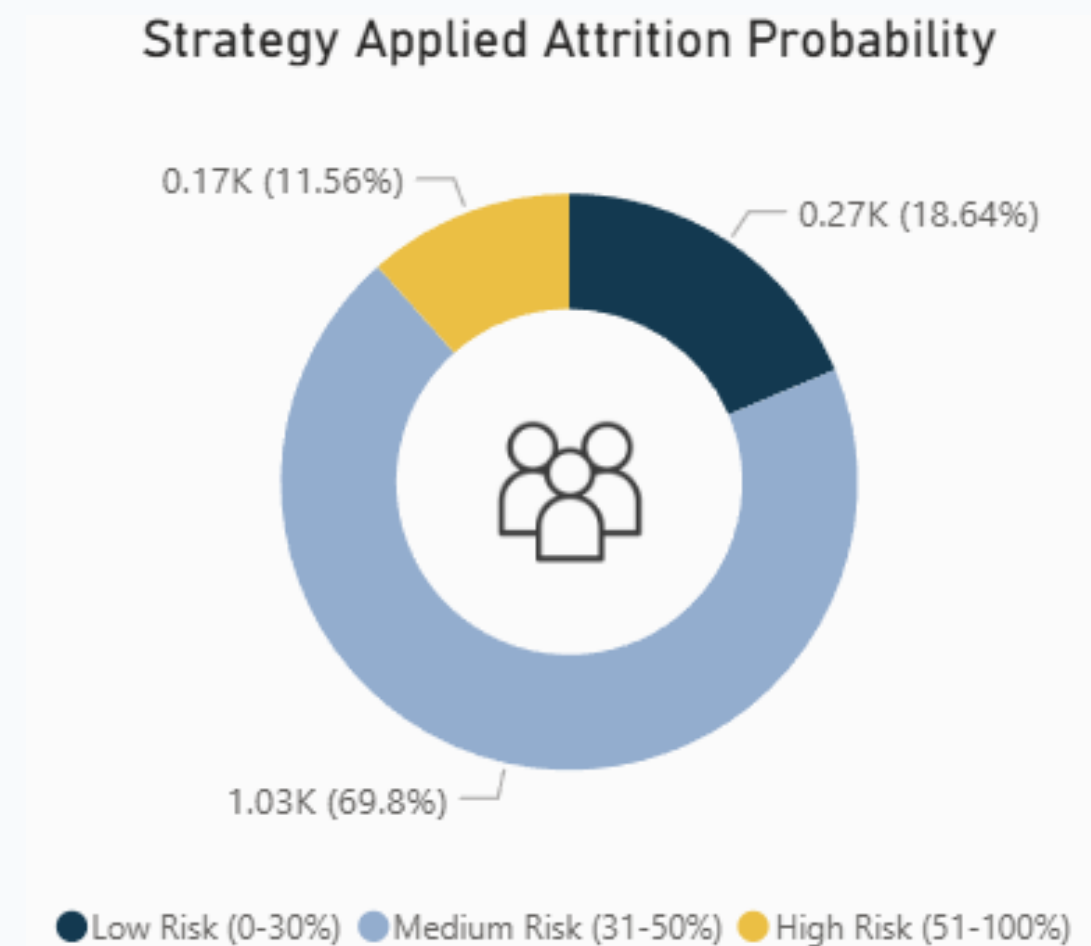
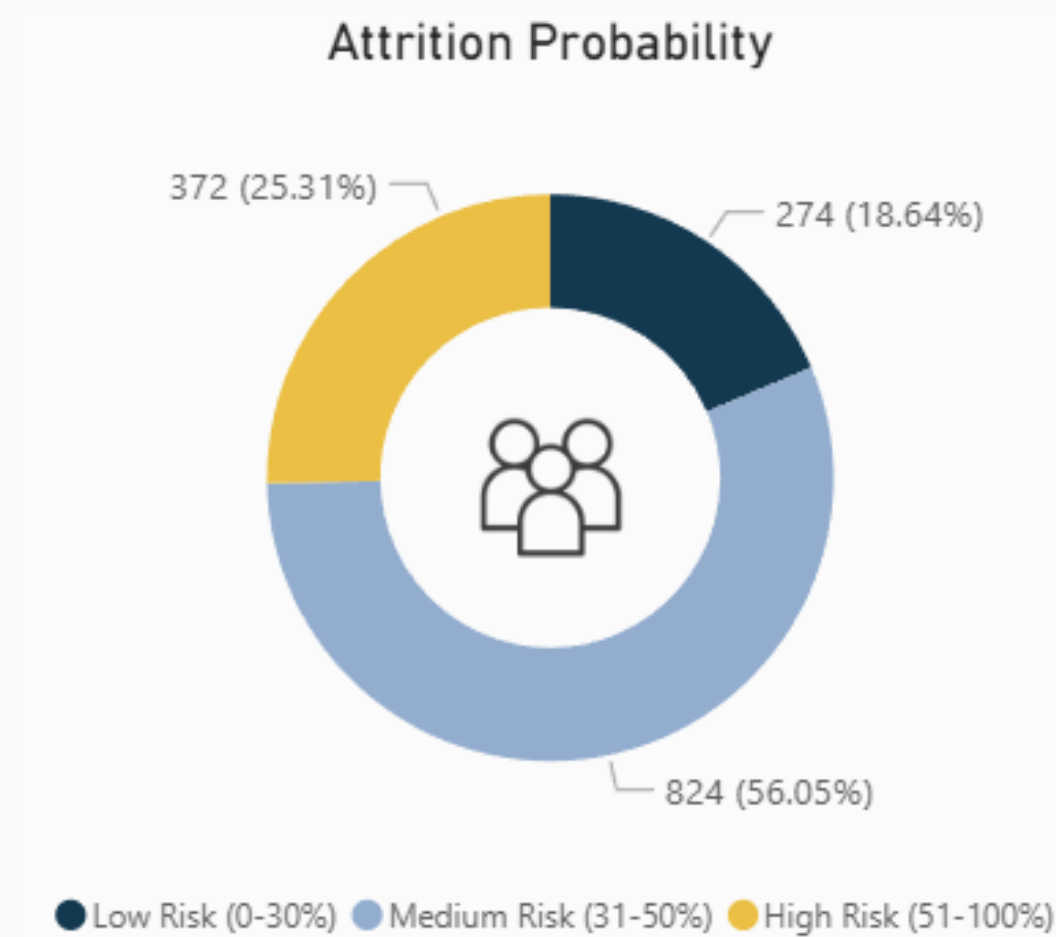
Factors	Strategy
Overtime	Restrict overtime for juniors
Salary	Increase salary based on performance: • Performance 1 = (increase icon) 2% • Performance 2 = (increase icon) 3% • Performance 3 = (increase icon) 4% • Performance 4 = (increase icon) 5%
	Increase salary: • Performance 4 + Job Involvement 3 = (increase icon) 10% • Performance 4 + Job Involvement 4 = (increase icon) 15%

Impact Analysis

Strategy Attrition Analysis

The prediction model after strategy is based on Logistic Regression.

Total employee: 1470



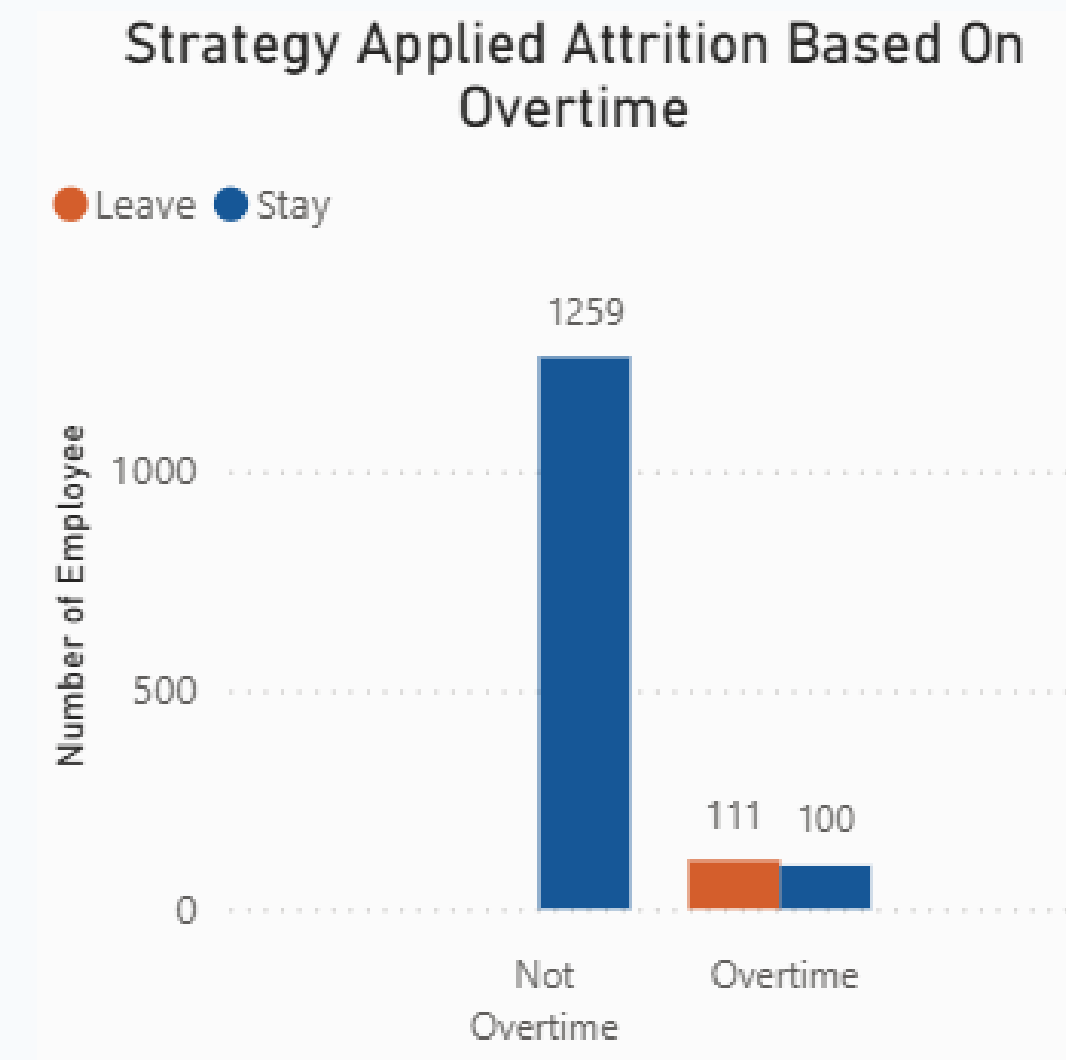
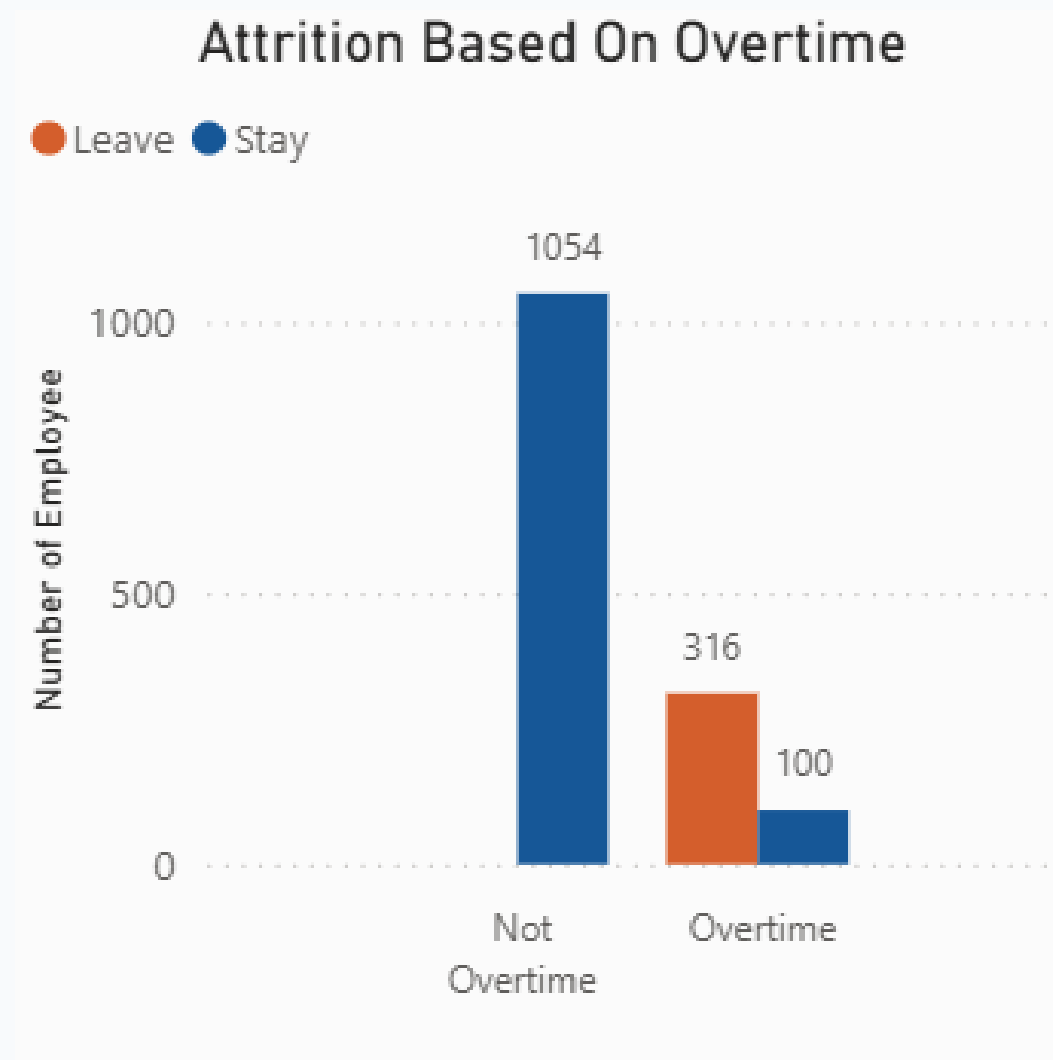
By implementing targeted retention strategies,

- Employees who have **high risk** (51%-100%) to leave the company decreased from **25.31%** to **11.56%** (↓ 13.75%)
- Employees who have **low risk** (0%-30%) to leave company increased from **18.64%** to **18.64%** (↑ 0%)

Strategy Attrition Analysis

The prediction model after strategy is based on Logistic Regression.

Total employee: 1470



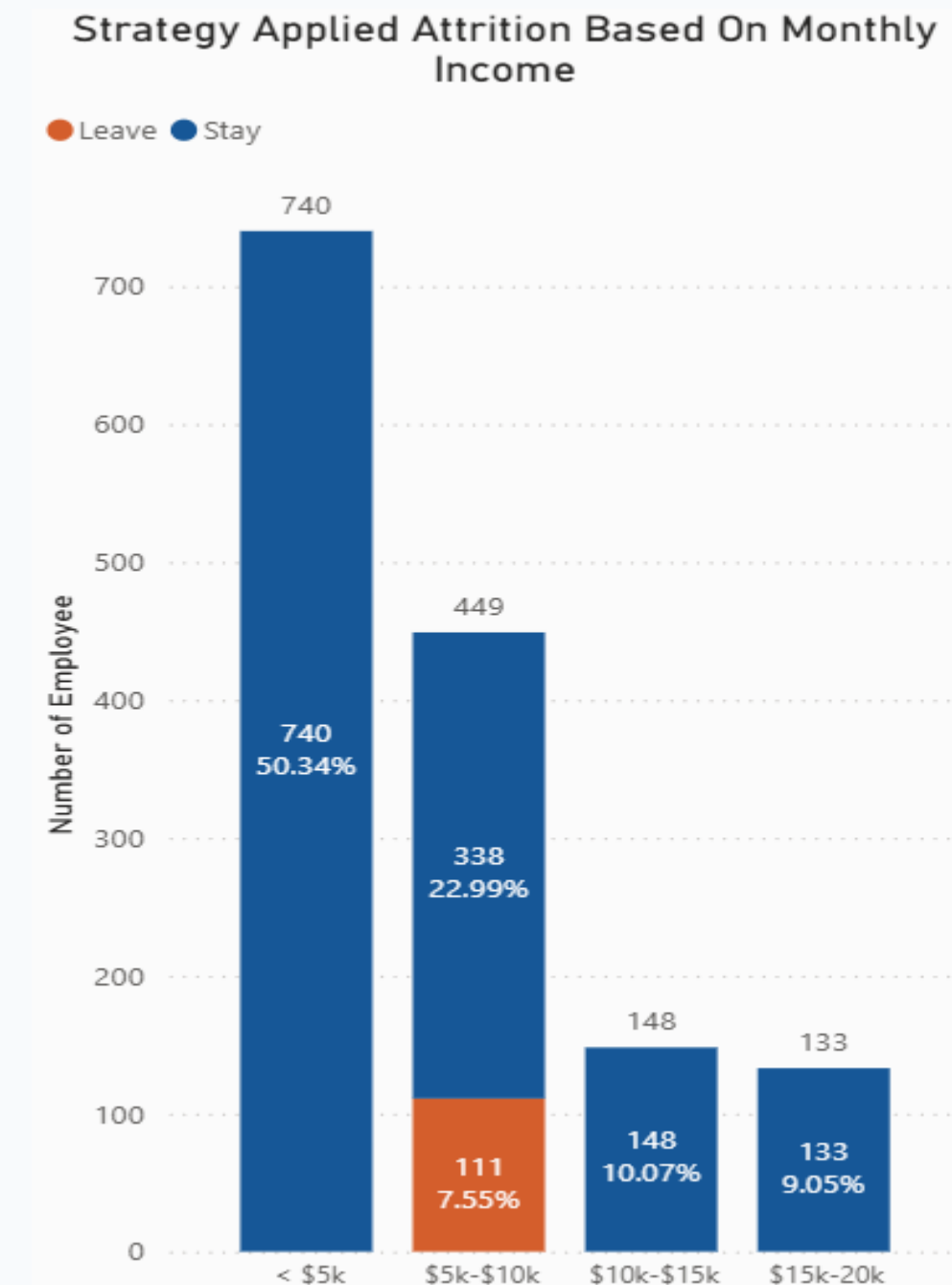
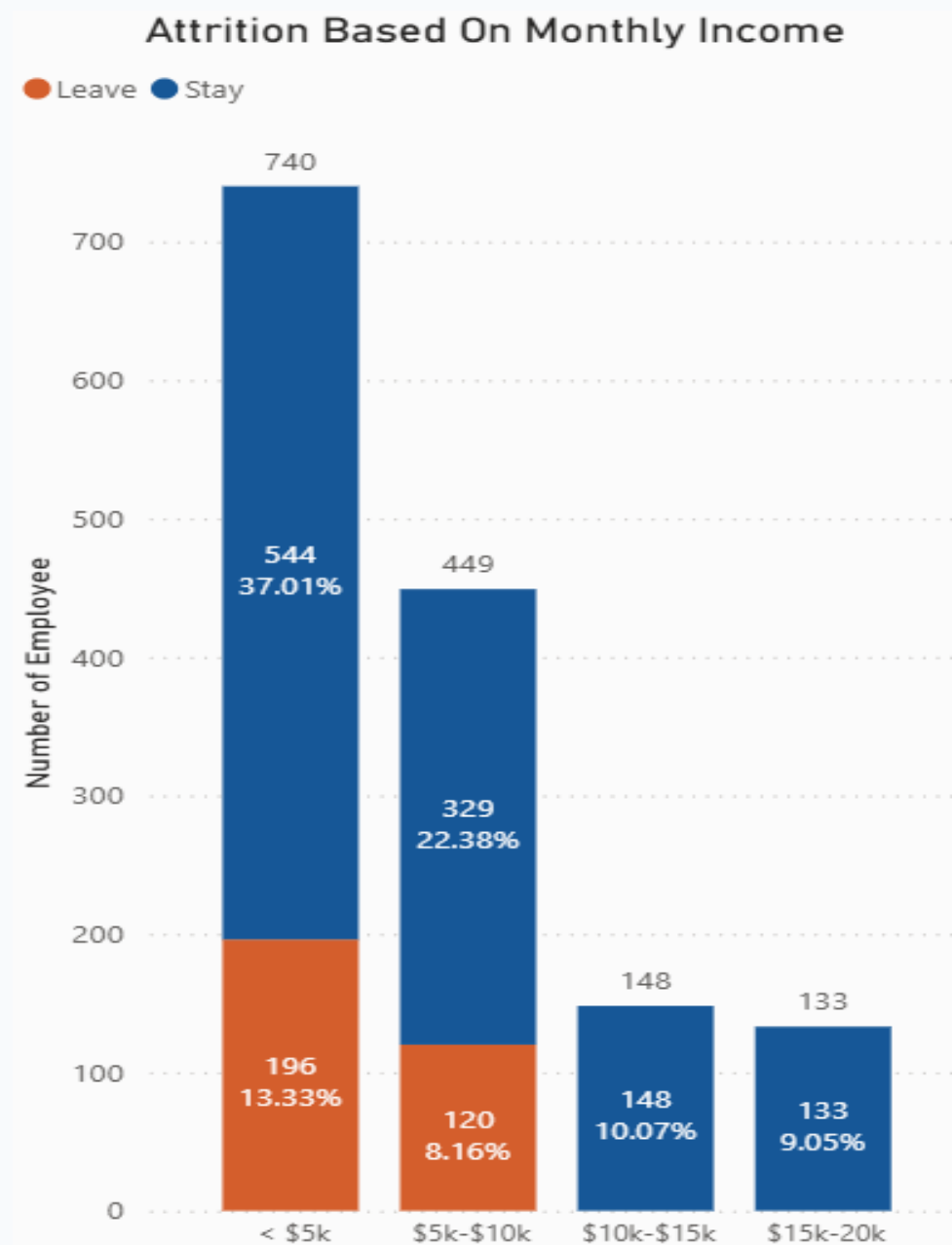
By implementing targeted retention strategies,

- Number of employees predicted to leave the company who **work overtime** is decreased from **316** to **111** (↓ 205)
- **None** of the employees who didn't work overtime predicted to leave the company.

Strategy Attrition Analysis

The prediction model after strategy is based on Logistic Regression.

Total employee: 1470



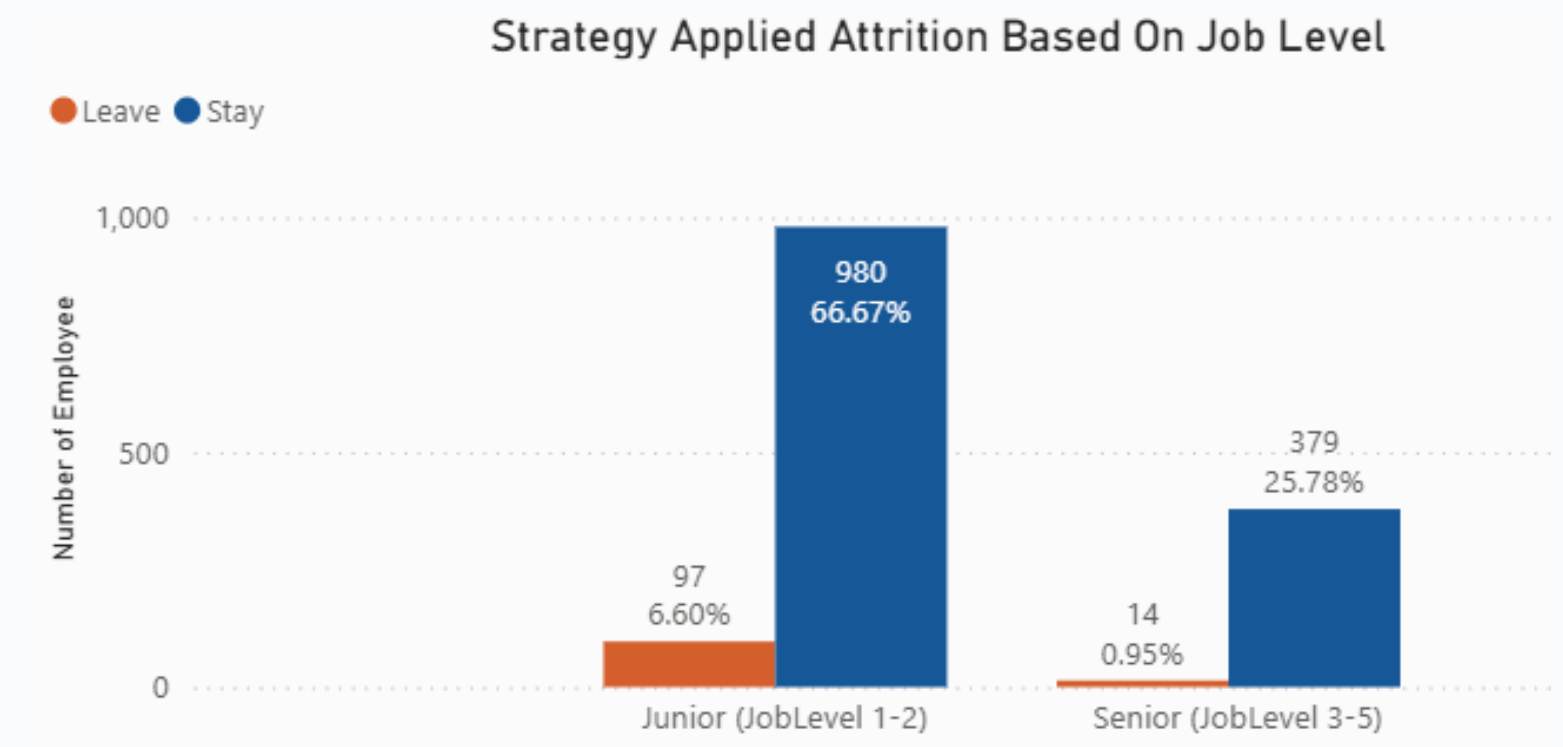
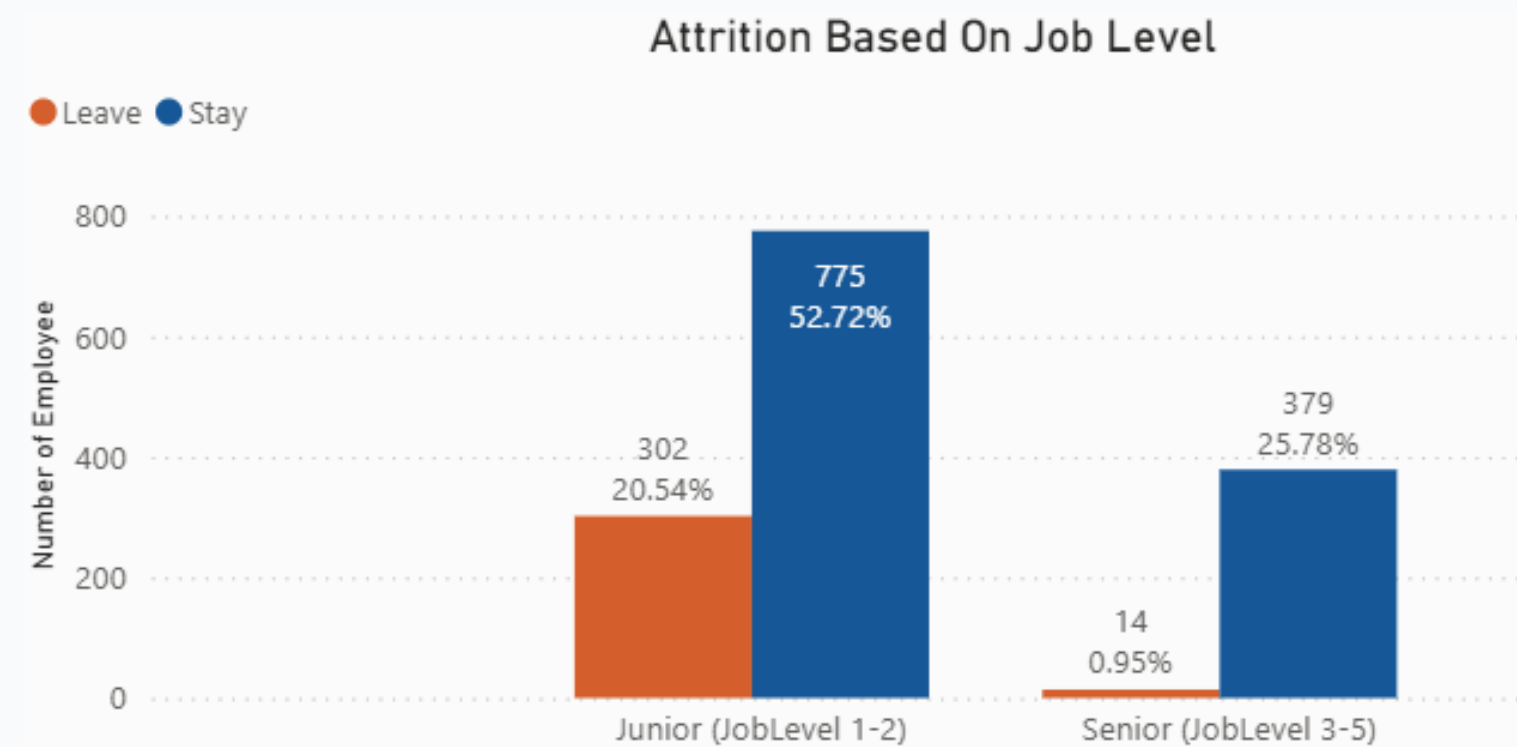
By implementing targeted retention strategies,

- Number of employee who predicted to leave company salary < 5000 decrease from 13.33% to 0%
- Number of employee who predicted to leave company salary between 5000 and 10,000 decrease from 8.16% to 7.55%

Strategy Attrition Analysis

The prediction model after strategy is based on Logistic Regression.

Total employee: 1470



By implementing targeted retention strategies,

- Number of **junior employees** predicted to leave the company from **302** to **97**
- Overall junior attrition rate from **20.54%** to **6.60%**

IBM EMPLOYEE ATTRITION PREDICTION DASHBOARD

Overtime Status

All

Salary Range

All

Job Level

All

Risk Range

All

Total Employee

1.47K

Average Income

6.50K

Attrition Rate

21.5%

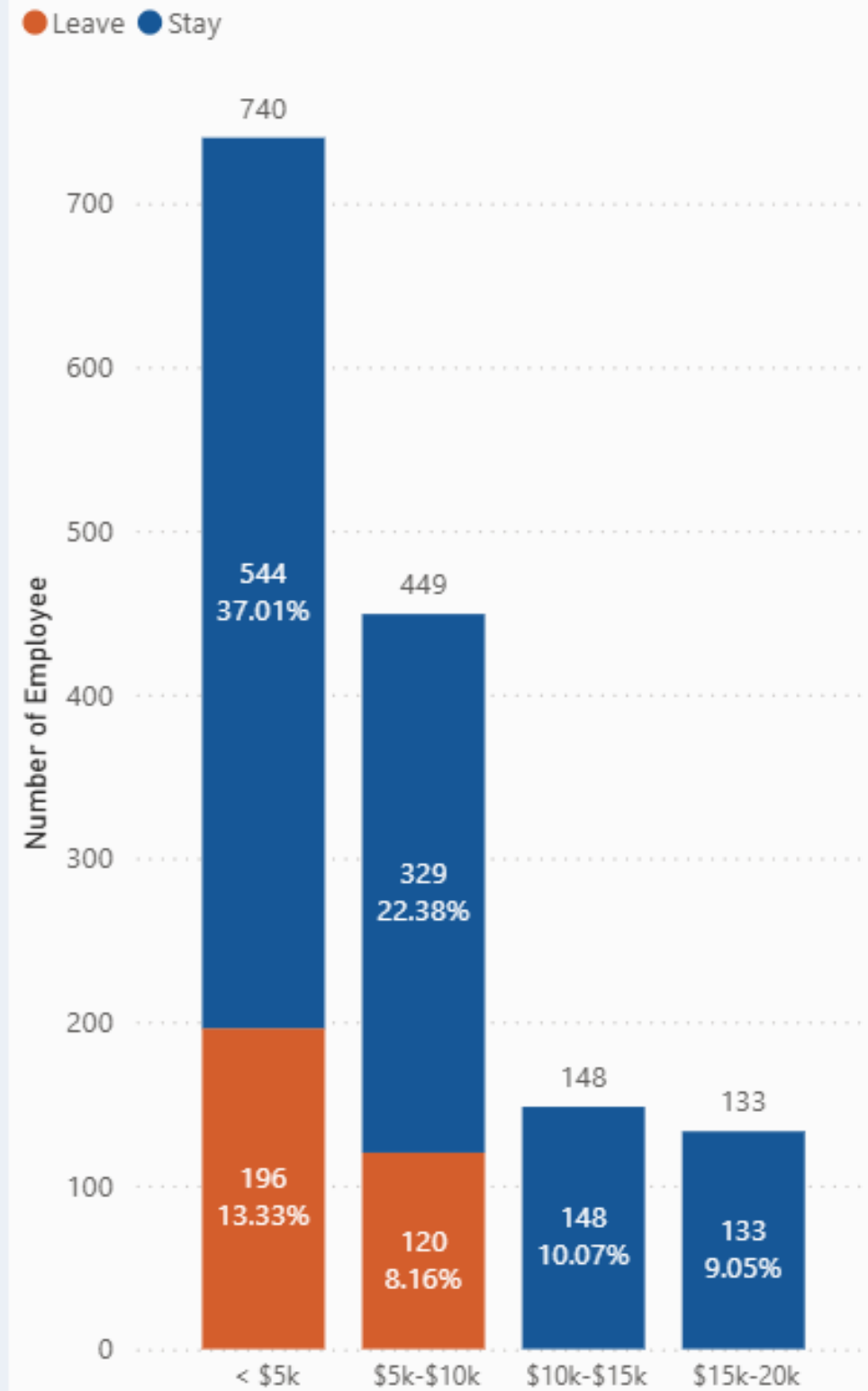
Junior Attrition

20.5%

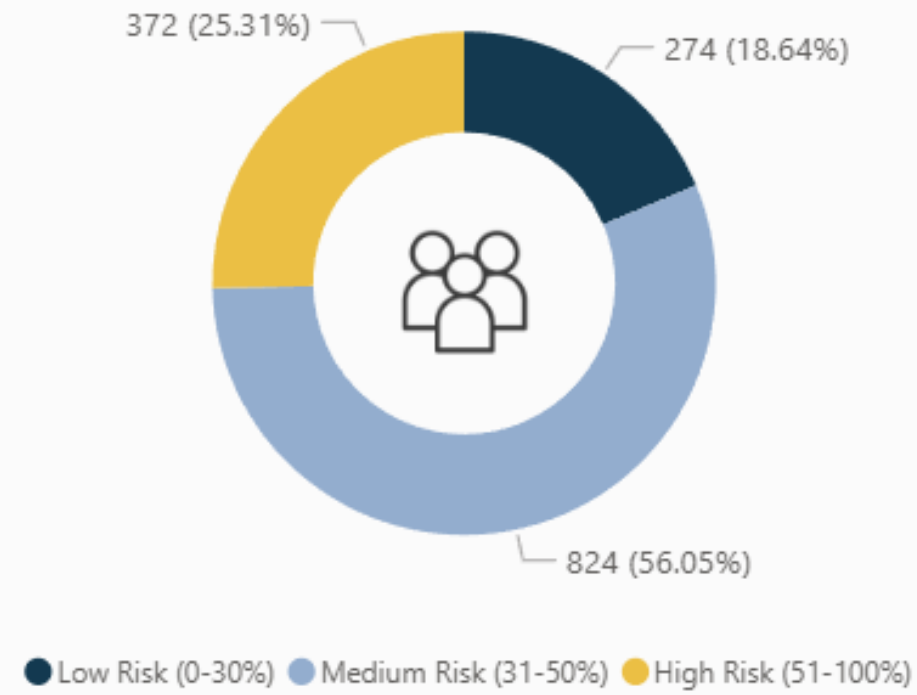
Total Overtime

237

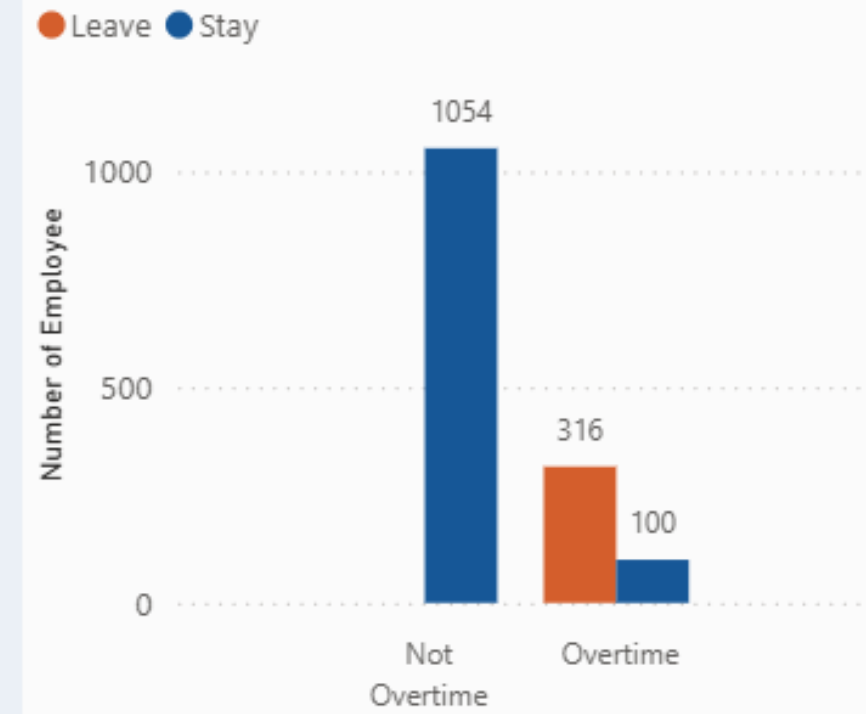
Attrition Based On Monthly Income



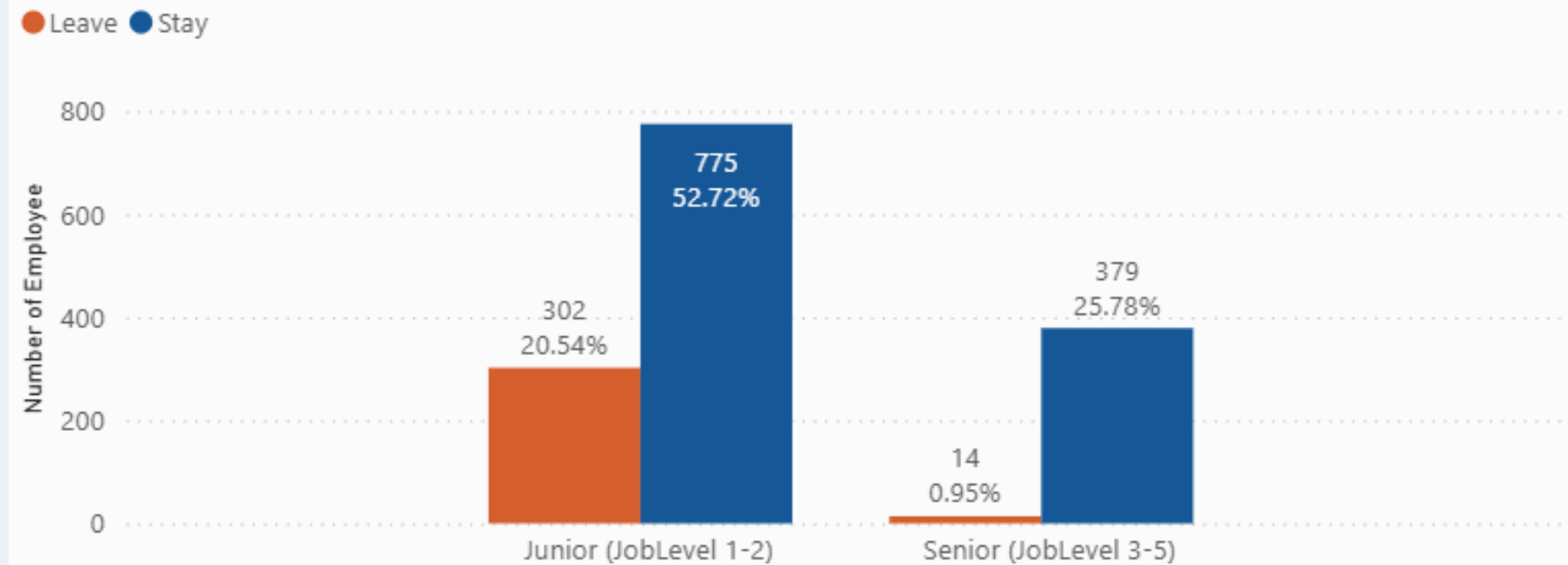
Attrition Probability



Attrition Based On Overtime



Attrition Based On Job Level



IBM EMPLOYEE ATTRITION STRATEGY DASHBOARD

Overtime Status: All | Salary Range: All | Job Level: All | Risk Range: All

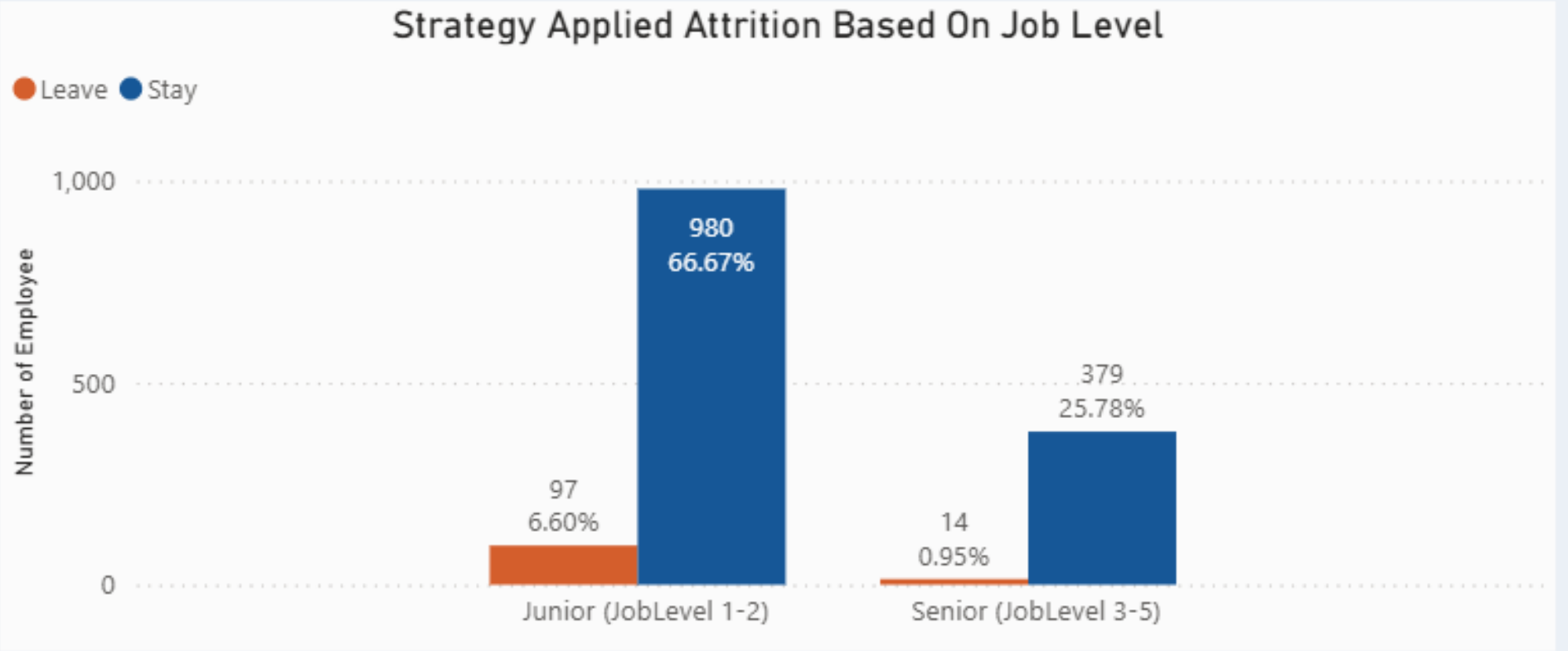
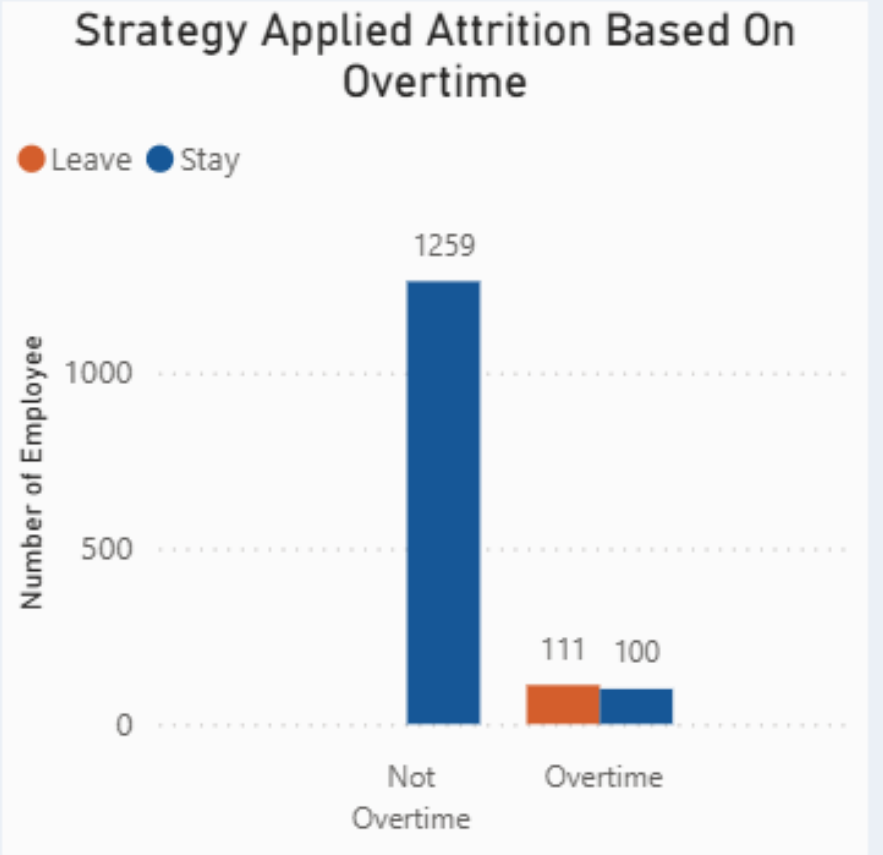
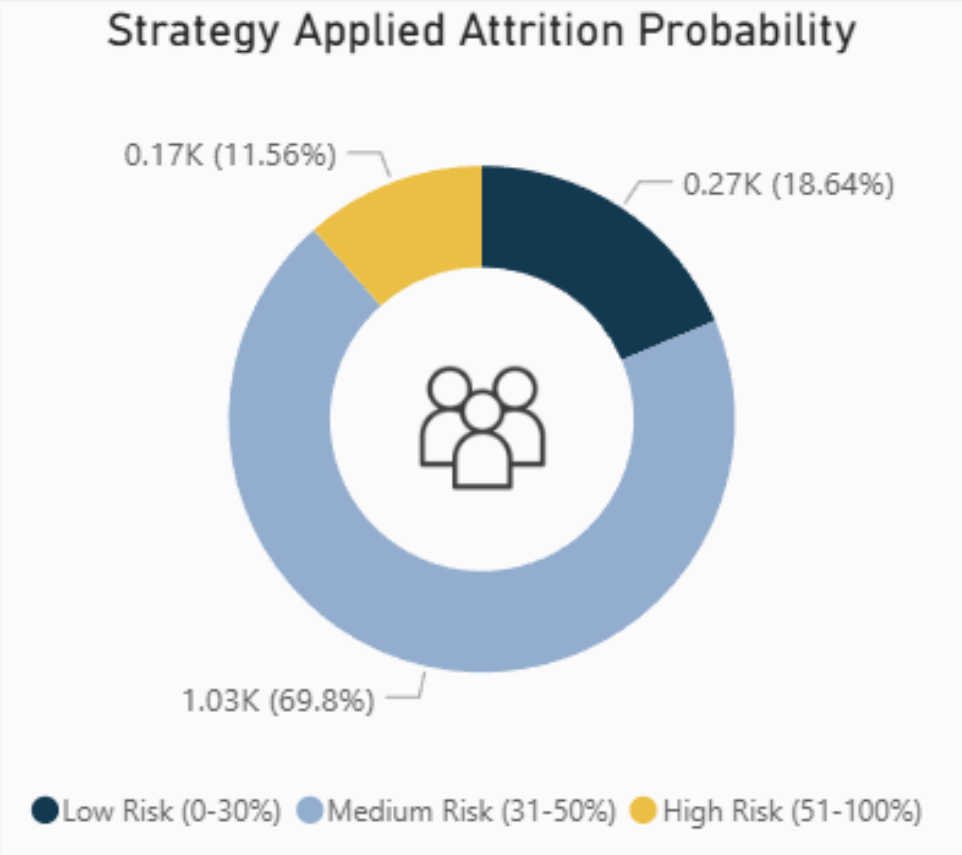
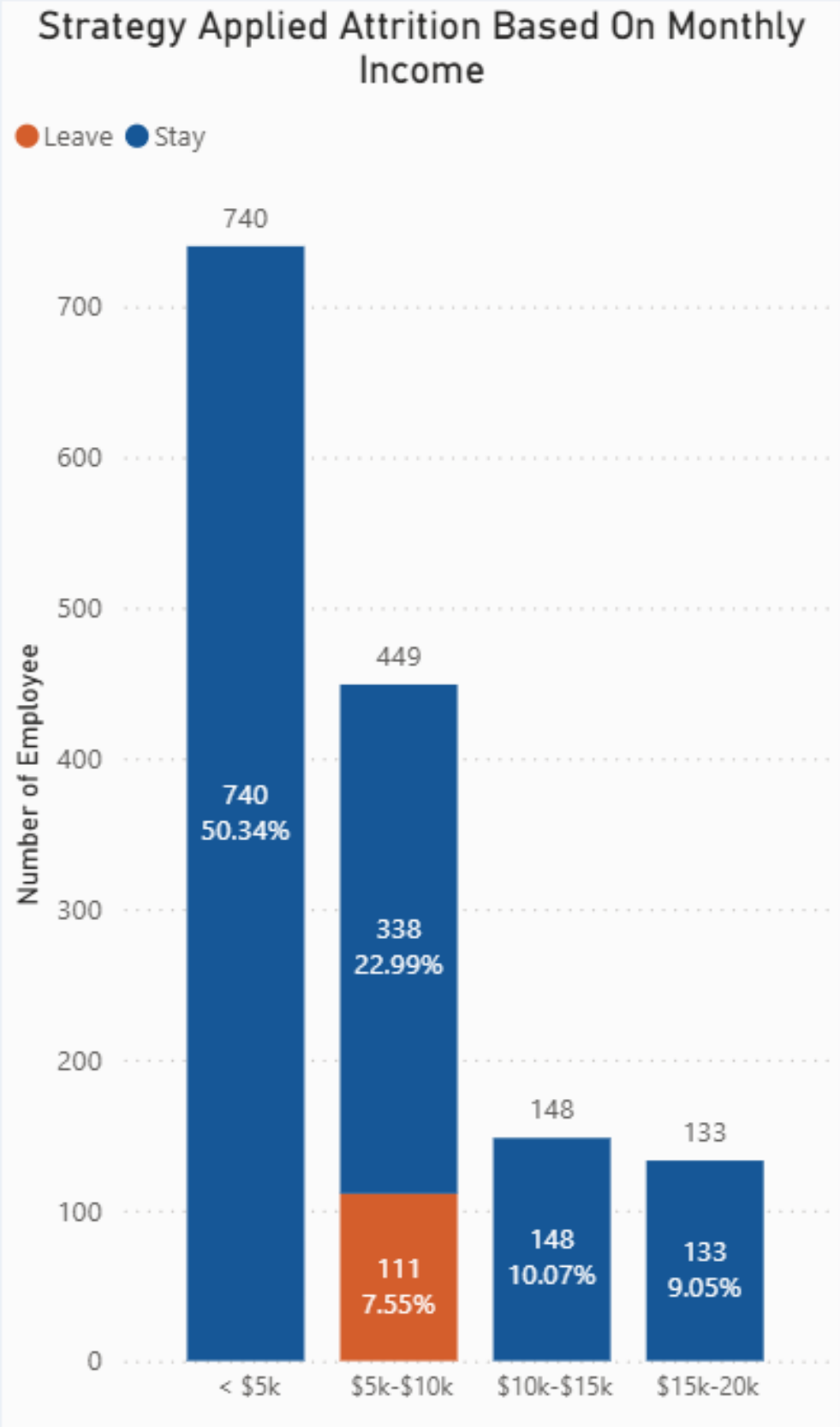
Total Employee
1.47K

Average Income
6.52K

Attrition Rate
7.6%

Junior Attrition
6.6%

Total OverTime
211



Business Insights

Insights & Recommendations

Projected Impact: Reducing Attrition from 16.1% to **7.6%**

Our predictive model reveals that attrition is not random. It is highly concentrated among junior staff facing excessive overtime and low pay.

Proactive interventions targeting these risks reduce turnover from 16.1% to 7.6%, surpassing our goal.

Recommendations:

- Systematically monitor and prioritize employees eligible for promotion based on their performance and job involvement to ensure long-term retention.
- Restrict overtime for junior staff to essential, urgent issues only, preventing burnout and reducing operational friction.



APPENDICES

| Dataset & Analysis Repository

[Dataset](#)

[Python](#)

[SQL](#)

[Dashboard](#)



THANK YOU
